AI-POWERED FINANCIAL REPORTING & BUDGET VARIANCE DASHBOARD

Prepared by

Afifah Khatoon

B.Com (NEP-2020), Belpahar College
Sambalpur University
Belpahar, Odisha

Acknowledgement

This project was made by me on my own. It is not a college project — it was not assigned, guided, or evaluated as part of any course at Belpahar College or Sambalpur University. I took it up independently, out of my own interest, as a self-directed portfolio project to build and demonstrate practical skills in financial reporting, data analytics, and Power BI.

1. Project Overview

This is a self-initiated portfolio project that builds an AI-powered financial reporting and budget variance dashboard. It brings together transactional actual-spend data (structured as it would be exported from Tally Prime), departmental budget targets maintained in Excel, and AI-generated commentary into a single analytics pipeline. The raw actual-vs-budget data (~10,000 records, 2021–2023) is cleaned and modelled into a star schema in Excel, variance is calculated using DAX measures in Power BI, and an AI insight layer flags the largest variances in an Excel dataset (variance_insights), with Google Gemini used to turn them into plain-language explanations — replacing the manual, line-by-line commentary a finance team would otherwise write by hand.

2. Scope of Project

In scope:

- **Data Cleaning & Staging:** Cleaning and staging ~10,000 transaction-level Actual vs Budget records (2021–2023) across 6 departments, 6 expense categories, and 5 regions.
- **Star-Schema Data Modeling:** Building a star-schema data model (Fact_Actuals, Fact_Budget, and supporting dimension tables: Dim_Department, Dim_Category, Dim_Region, Dim_Date) in Excel, ready for Power BI import.
- **ERP Data Lineage Documentation:** Documenting the real-world data source

methodology: exporting the Day Book / General Ledger from Tally Prime.

- **Variance Calculations:** Calculating variance (₹ and %) using DAX measures at Department, Category, and monthly-trend levels.
- **AI Insight Layer:** Building an Excel-based AI insight layer (variance_insights) that flags variances above 15%, and using Google Gemini to generate plain-language narrative commentary, presented in this report's AI-Generated Variance Insights section.
- **Interactive Power BI Dashboard:** Designing an interactive Power BI dashboard layout (KPI cards, yearly Actual vs Budget trend, departmental spend breakdown by donut chart, and monthly variance trend line chart by department).

Out of scope:

- **Live ERP Integration:** Live/real-time integration with a production Tally Prime instance (a manual export is used to simulate this).
- **Multi-Currency Consolidation:** Multi-company or multi-currency consolidation (all amounts are in ₹ INR).
- **Predictive Budgeting:** Forecasting or predictive budgeting (the project covers variance of actuals against an existing budget, not budget-setting itself).

3. Project Objective

- Automate the actual-vs-budget variance calculation that is otherwise done manually in spreadsheets
- Build a clean, reusable star-schema data model (fact and dimension tables) that can absorb new months and departments without rework
- Surface over-budget and under-budget areas quickly through an interactive Power BI dashboard (KPI cards, yearly trend, departmental breakdown, and monthly variance trend)
- Reduce the time spent writing variance commentary by flagging key variances in an Excel insight layer and using Google Gemini to generate first-draft, plain-language explanations
- Demonstrate an end-to-end, real-world accounting-to-analytics pipeline (Tally Prime → Excel → Power Query → Power BI, with AI-generated insights) as a self-made portfolio project for Data Analyst / Business Analyst / MIS Executive roles

4. Tools & Knowledge Used

- **Tally Prime:** Source of actual transaction data (Day Book / General Ledger export).
- **Microsoft Excel:** Data cleaning, star-schema staging, variance formulas (in ₹), and KPI summary sheets.
- **Power Query:** ETL processing—cleaning, de-duplication, and mapping raw data into fact and dimension tables.
- **Power BI:** Data modeling (star schema), relationship management, and interactive dashboard visuals (KPI cards, bar, donut, and line charts).

- **DAX:** Variance measures—Total Actual, Total Budget, Variance (₹), Variance (%), YTD, and MoM trend calculations.
- **AI Insights (Excel + Google Gemini):** Excel-based anomaly flagging (variance_insights) and Google Gemini for plain-language variance commentary.
- **Data Modeling Concepts:** Star schema design, fact/dimension table architecture, surrogate keys, and data quality flagging.
- **Accounting Knowledge:** General ledger analysis, voucher classification, departmental budgeting, and variance analysis.

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5. Methodology / Approach

Step 1: Extract: Export actual transactions from Tally Prime (Day Book / General Ledger); gather departmental budget targets in Excel.

Step 2: Transform: Clean, de-duplicate, and flag data quality issues; map to a star schema (Fact_Actuals, Fact_Budget, Dim_Date, Dim_Department, Dim_Category, Dim_Region).

Step 3: Model: Load the star schema into Power BI and build relationships.

Step 4: Calculate: Write DAX measures for variance (₹ and %), status flags, and month-over-month trend.

Step 5: Generate insight: Flag variances above 15% in the variance_insights dataset and use Google Gemini to write plain-language commentary.

Step 6: Visualize: Build an interactive Power BI dashboard with KPI cards, a yearly Actual vs Budget chart, a department donut chart, and a monthly variance trend line chart.

6. Expected Outcome

A working, interactive Power BI dashboard that shows, at a glance, where actual spend is running over or under budget by year, department, and month — paired with automatically generated, plain-language commentary (in the AI-Generated Variance Insights section) explaining the biggest variances, so a reviewer does not need to read raw numbers to understand what happened and where to focus attention.

Dataset Information:

Dataset Name: Budget_vs_Actual_Data

DATASET LINK: https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Budget_vs_Actual_Data.xlsx

Chart of Accounts & Master Ledger Configuration In Tally Prime:

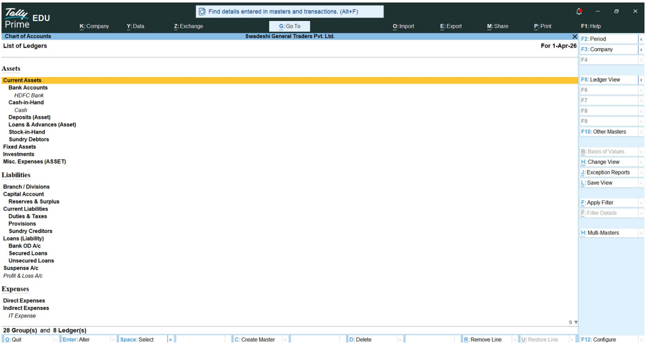


Chart of Accounts — Swadeshi General Traders Pvt. Ltd.

List of Ledgers — For 1-Apr-26

Assets

- Current Assets
  - Bank Accounts
    - *HDFC Bank*
  - Cash-in-Hand
    - *Cash*
  - Deposits (Asset)
  - Loans & Advances (Asset)
  - Stock-in-Hand
  - Sundry Debtors
- Fixed Assets
- Investments
- Misc. Expenses (ASSET)

Liabilities

- Branch / Divisions
- Capital Account
  - Reserves & Surplus
- Current Liabilities
  - Duties & Taxes
  - Provisions
  - Sundry Creditors
- Loans (Liability)
  - Bank OD A/c
  - Secured Loans
  - Unsecured Loans
- Suspense A/c
- *Profit & Loss A/c*

Expenses

- Direct Expenses
- Indirect Expenses
  - *IT Expense*

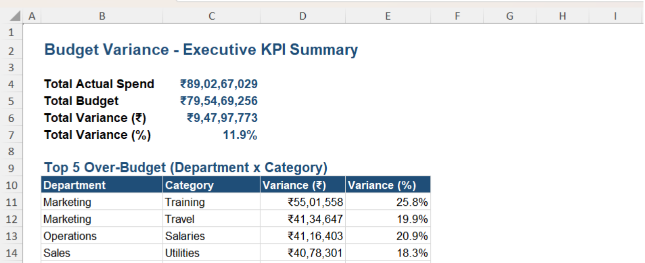
28 Group(s) and 8 Ledger(s)

Explanation:

To establish a direct link between transactional accounting and downstream data analytics, the ERP foundation was configured in Tally Prime under the entity Swadeshi General Traders Pvt. Ltd.:

- Chart of Accounts Setup: Verified the baseline ledger architecture across primary financial groups including Assets (Bank & Cash Accounts), Liabilities, and Expenses (Indirect Expenses).
- ERP Data Origin: Ledgers such as *HDFC Bank* and *IT Expense* serve as the transactional source endpoints for logging operational expenditures prior to exporting General Ledger records for ETL staging in Microsoft Excel and visualization in Power BI.

Data Staging, Star-Schema Architecture, and Executive KPI Summary in Microsoft Excel:

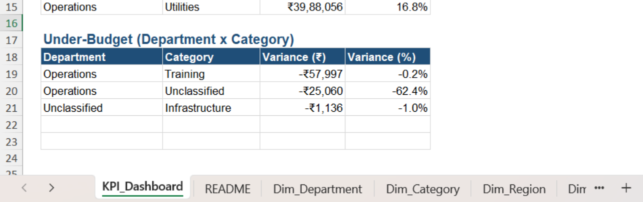


Budget Variance - Executive KPI Summary

Total Actual Spend	₹89,02,67,029
Total Budget	₹79,54,69,256
Total Variance (₹)	₹9,47,97,773
Total Variance (%)	11.9%

Top 5 Over-Budget (Department x Category)

Department	Category	Variance (₹)	Variance (%)
Marketing	Training	₹55,01,558	25.8%
Marketing	Travel	₹41,34,647	19.9%
Operations	Salaries	₹41,16,403	20.9%
Sales	Utilities	₹40,78,301	18.3%



Operations	Utilities	₹39,88,056	16.8%

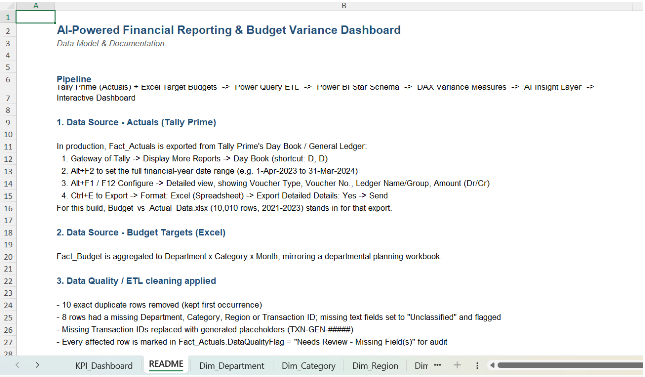
Under-Budget (Department x Category)

Department	Category	Variance (₹)	Variance (%)
Operations	Training	-₹57,997	-0.2%
Operations	Unclassified	-₹25,060	-62.4%
Unclassified	Infrastructure	-₹1,136	-1.0%

Explanation: Prior to Power BI ingestion, the data model was staged and validated in Microsoft Excel under a structured Star Schema (Fact_Actuals, Fact_Budget, and supporting Dim_* sheets):

- **Executive Baseline Reconciliation:** The primary financial metrics were reconciled to establish overall baseline benchmarks. Total Actual Spend of ₹89,02,67,029 against a Total Budget of ₹79,54,69,256 resulted in a net over-budget variance of ₹9,47,97,773 (+11.9%).
- **Variance Driver Isolation:** The KPI summary isolates the top 5 over-budget Department × Category combinations, led by Marketing Training (+25.8% / ₹55,01,558), followed by Marketing Travel, Operations Salaries, Sales Utilities, and Operations Utilities. It also lists the 3 combinations that ran under budget: Operations Training (-₹57,997), Operations Unclassified (-₹25,060 / -62.4%), and Unclassified Infrastructure (-₹1,136).

Data Lineage, Extraction Methodology, and Power Query Data Cleaning Audit (README):



AI-Powered Financial Reporting & Budget Variance Dashboard

Data Model & Documentation

Pipeline

Tally Prime (Actuals) + Excel Target Budgets -> Power Query ETL -> Power BI Star Schema -> DAX Variance Measures -> AI Insight Layer -> Interactive Dashboard

1. Data Source - Actuals (Tally Prime)

In production, Fact_Actuals is exported from Tally Prime's Day Book / General Ledger:
1. Gateway of Tally -> Display More Reports -> Day Book (shortcut: D, D)
2. Alt+F2 to set the full financial-year date range (e.g. 1-Apr-2023 to 31-Mar-2024)
3. Alt+F1 / F12 Configure -> Detailed view, showing Voucher Type, Voucher No., Ledger Name/Group, Amount (Dr/Cr)
4. Ctrl+E to Export -> Format: Excel (Spreadsheet) -> Export Detailed Details: Yes -> Send

For this build, Budget_vs_Actual_Data.xlsx (10,010 rows, 2021-2023) stands in for that export.

2. Data Source - Budget Targets (Excel)

Fact_Budget is aggregated to Department x Category x Month, mirroring a departmental planning workbook.

3. Data Quality / ETL cleaning applied

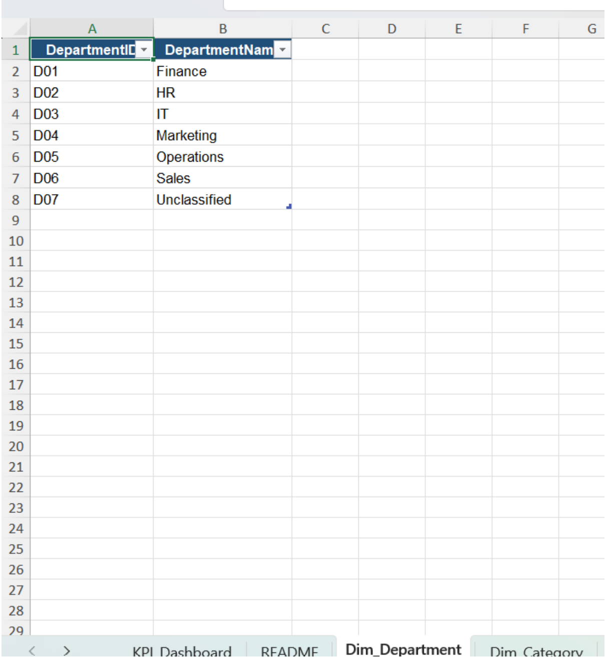
- 10 exact duplicate rows removed (kept first occurrence)
- 8 rows had a missing Department, Category, Region or Transaction ID; missing text fields set to "Unclassified" and flagged
- Missing Transaction IDs replaced with generated placeholders (TXN-GEN-#####)
- Every affected row is marked in Fact_Actuals.DataQualityFlag = "Needs Review - Missing Field(s)" for audit

Explanation:

The **README** sheet establishes the official technical data lineage, ERP extraction workflow, and ETL auditing protocols for the financial reporting pipeline:

- **Tally Prime General Ledger Extraction Protocol:** Documents the step-by-step Tally export procedure (Gateway of Tally Day Book Alt+F2 Date Range Ctrl+E Excel Export) used to mirror production accounting data flows.
- **Power Query Data Cleaning & Quality Control:** Outlines the specific ETL transformations applied to the 10,010-row dataset, including de-duplication (10 exact duplicate rows removed), handling missing fields by setting null text attributes to *"Unclassified"*, generating placeholder transaction keys (TXN-GEN-#####), and tagging impacted records with DataQualityFlag = "Needs Review - Missing Field(s)" for auditability.

Normalized Department Dimension Table (Dim_Department):



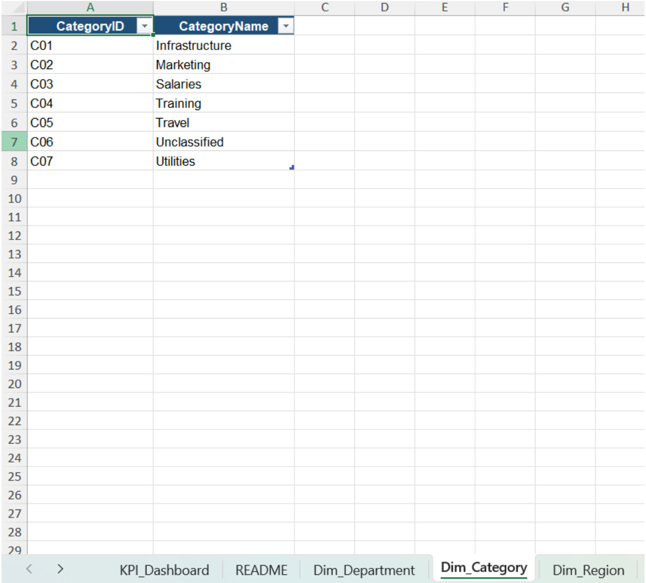
DepartmentID	DepartmentNam
D01	Finance
D02	HR
D03	IT
D04	Marketing
D05	Operations
D06	Sales
D07	Unclassified

Explanation:

The **Dim_Department** lookup table establishes a clean primary key structure for organizational attributes within the Star Schema:

- **Attribute Mapping:** Maps unique primary keys (DepartmentID from D01 to D07) to standardized business units including Finance, HR, IT, Marketing, Operations, Sales, and *Unclassified*.
- **Relational Integrity:** Connects as a primary lookup table to both Fact_Actuals and Fact_Budget tables, enabling cross-departmental filtering and drill-down analysis without data duplication.

Expense Category Dimension Table (Dim_Category):



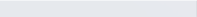
CategoryID	CategoryName
C01	Infrastructure
C02	Marketing
C03	Salaries
C04	Training
C05	Travel
C06	Unclassified
C07	Utilities

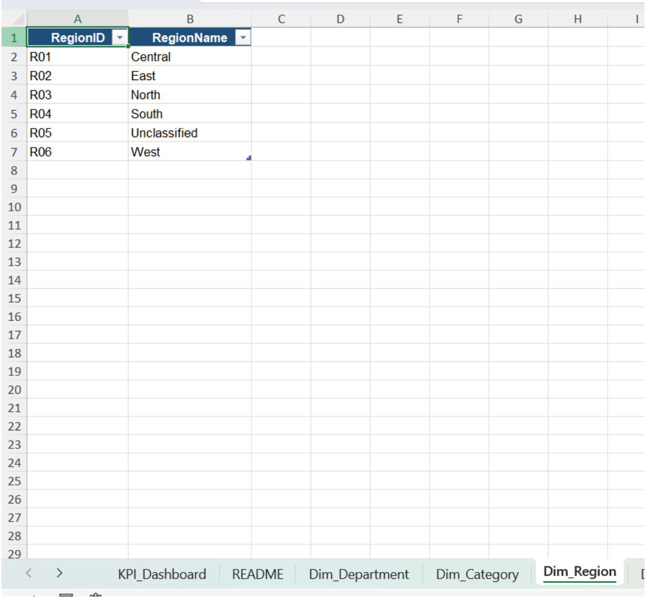
Explanation:

The **Dim_Category** dimension table normalizes expense classifications across the data model to ensure consistent financial reporting:

- **Category Mapping:** Maps primary keys (CategoryID from C01 to C07) to standardized operational cost heads including Infrastructure, Marketing, Salaries, Training, Travel, Unclassified, and Utilities.
- **Schema Integration:** Acts as a primary lookup table connecting via relationships to both Fact_Actuals and Fact_Budget tables, enabling granular line-item expense analysis and variance slicing across all business units.

Geographic Region Dimension Table (Dim_Region):





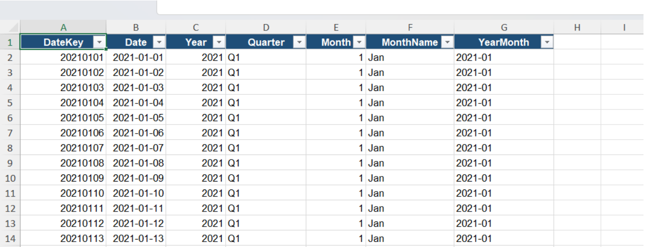
RegionID	RegionName
R01	Central
R02	East
R03	North
R04	South
R05	Unclassified
R06	West

Explanation:

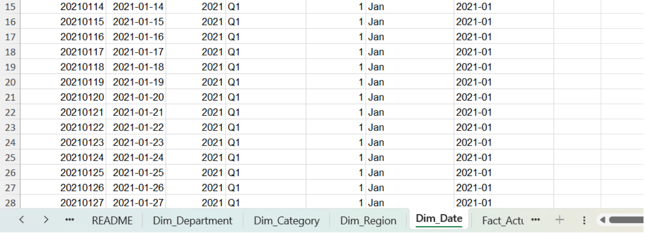
The Dim_Region lookup table provides normalized spatial attributes to support regional performance slicing across the dataset:

- Geographic Mapping: Maps primary keys (RegionID from R01 to R06) to operational zones including Central, East, North, South, West, and *Unclassified*.
- Relational Schema Integration: Connects as a primary lookup table to Fact_Actuals, allowing multi-dimensional filtering, regional budget variance analysis, and geographical breakdown within Power BI reports.

Calendar Master Dimension Table (Dim_Date):



DateKey	Date	Year	Quarter	Month	MonthName	YearMonth
20210101	2021-01-01	2021	Q1	1	Jan	2021-01
20210102	2021-01-02	2021	Q1	1	Jan	2021-01
20210103	2021-01-03	2021	Q1	1	Jan	2021-01
20210104	2021-01-04	2021	Q1	1	Jan	2021-01
20210105	2021-01-05	2021	Q1	1	Jan	2021-01
20210106	2021-01-06	2021	Q1	1	Jan	2021-01
20210107	2021-01-07	2021	Q1	1	Jan	2021-01
20210108	2021-01-08	2021	Q1	1	Jan	2021-01
20210109	2021-01-09	2021	Q1	1	Jan	2021-01
20210110	2021-01-10	2021	Q1	1	Jan	2021-01
20210111	2021-01-11	2021	Q1	1	Jan	2021-01
20210112	2021-01-12	2021	Q1	1	Jan	2021-01
20210113	2021-01-13	2021	Q1	1	Jan	2021-01



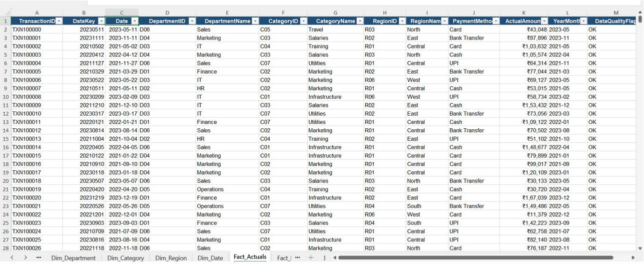
15	20210114	2021-01-14	2021	Q1	1	Jan	2021-01
16	20210115	2021-01-15	2021	Q1	1	Jan	2021-01
17	20210116	2021-01-16	2021	Q1	1	Jan	2021-01
18	20210117	2021-01-17	2021	Q1	1	Jan	2021-01
19	20210118	2021-01-18	2021	Q1	1	Jan	2021-01
20	20210119	2021-01-19	2021	Q1	1	Jan	2021-01
21	20210120	2021-01-20	2021	Q1	1	Jan	2021-01
22	20210121	2021-01-21	2021	Q1	1	Jan	2021-01
23	20210122	2021-01-22	2021	Q1	1	Jan	2021-01
24	20210123	2021-01-23	2021	Q1	1	Jan	2021-01
25	20210124	2021-01-24	2021	Q1	1	Jan	2021-01
26	20210125	2021-01-25	2021	Q1	1	Jan	2021-01
27	20210126	2021-01-26	2021	Q1	1	Jan	2021-01
28	20210127	2021-01-27	2021	Q1	1	Jan	2021-01

Explanation:

The **Dim_Date** dimension table serves as the central date master in the Star Schema to support time-intelligence analytics, trend modeling, and period-over-period financial evaluations:

- **Surrogate Key & Hierarchy Attributes:** Contains a numeric primary key (DateKey formatted as YYYYMMDD) alongside calendar attributes including Date, Year, Quarter, Month, MonthName, and YearMonth.
- **Time-Intelligence Support:** Connects via relationships to both Fact_Actuals (using DateKey) and Fact_Budget (using YearMonth), enabling seamless DAX time-intelligence calculations such as Year-to-Date (YTD) actual spend and Month-over-Month (MoM) budget variance tracking.

Actual Transactions Fact Table (Fact_Actuals):



	TransactionID	DateKey	Date	DepartmentID	DepartmentName	CategoryID	CategoryName	RegionID	RegionNam	PaymentMetho	ActualAmoun	YearMont	DataQualityFlag
2	TXN100000	20230511	2023-05-11	D06	Sales	C05	Travel	R03	North	Card	₹43,048	2023-05	OK
3	TXN100001	20231111	2023-11-11	D04	Marketing	C03	Salaries	R02	East	Bank Transfer	₹87,896	2023-11	OK
4	TXN100002	20210502	2021-05-02	D03	IT	C04	Training	R01	Central	Card	₹1,03,632	2021-05	OK
5	TXN100003	20220412	2022-04-12	D04	Marketing	C03	Salaries	R03	North	Cash	₹1,05,574	2022-04	OK
6	TXN100004	20211127	2021-11-27	D06	Sales	C07	Utilities	R01	Central	UPI	₹64,314	2021-11	OK
7	TXN100005	20210329	2021-03-29	D01	Finance	C02	Marketing	R02	East	Bank Transfer	₹77,044	2021-03	OK
8	TXN100006	20230522	2023-05-22	D03	IT	C02	Marketing	R06	West	UPI	₹69,127	2023-05	OK
9	TXN100007	20210511	2021-05-11	D02	HR	C02	Marketing	R01	Central	Cash	₹53,015	2021-05	OK
10	TXN100008	20230209	2023-02-09	D03	IT	C01	Infrastructure	R06	West	UPI	₹58,734	2023-02	OK
11	TXN100009	20211210	2021-12-10	D03	IT	C03	Salaries	R02	East	Cash	₹1,53,432	2021-12	OK
12	TXN100010	20230317	2023-03-17	D03	IT	C07	Utilities	R02	East	Bank Transfer	₹73,056	2023-03	OK
13	TXN100011	20220121	2022-01-21	D01	Finance	C07	Utilities	R01	Central	Cash	₹1,09,122	2022-01	OK
14	TXN100012	20230814	2023-08-14	D06	Sales	C02	Marketing	R01	Central	Bank Transfer	₹70,502	2023-08	OK
15	TXN100013	20211004	2021-10-04	D02	HR	C04	Training	R02	East	UPI	₹51,102	2021-10	OK
16	TXN100014	20220405	2022-04-05	D06	Sales	C01	Infrastructure	R01	Central	Cash	₹1,48,677	2022-04	OK
17	TXN100015	20210122	2021-01-22	D04	Marketing	C01	Infrastructure	R01	Central	Card	₹79,899	2021-01	OK
18	TXN100016	20210910	2021-09-10	D04	Marketing	C02	Marketing	R01	Central	Card	₹99,017	2021-09	OK
19	TXN100017	20230118	2023-01-18	D04	Marketing	C02	Marketing	R01	Central	Card	₹1,20,109	2023-01	OK
20	TXN100018	20230507	2023-05-07	D06	Sales	C03	Salaries	R03	North	Bank Transfer	₹30,133	2023-05	OK
21	TXN100019	20220420	2022-04-20	D05	Operations	C04	Training	R02	East	Cash	₹30,720	2022-04	OK
22	TXN100020	20231219	2023-12-19	D01	Finance	C01	Infrastructure	R02	East	Card	₹1,67,039	2023-12	OK
23	TXN100021	20220526	2022-05-26	D05	Operations	C07	Utilities	R04	South	Bank Transfer	₹1,49,486	2022-05	OK
24	TXN100022	20221201	2022-12-01	D04	Marketing	C02	Marketing	R06	West	Card	₹11,379	2022-12	OK
25	TXN100023	20230903	2023-09-03	D01	Finance	C03	Salaries	R04	South	UPI	₹1,42,223	2023-09	OK
26	TXN100024	20210709	2021-07-09	D06	Sales	C07	Utilities	R01	Central	UPI	₹62,758	2021-07	OK
27	TXN100025	20230816	2023-08-16	D04	Marketing	C01	Infrastructure	R01	Central	UPI	₹82,140	2023-08	OK
28	TXN100026	20221118	2022-11-18	D06	Sales	C02	Marketing	R03	North	Card	₹76,187	2022-11	OK

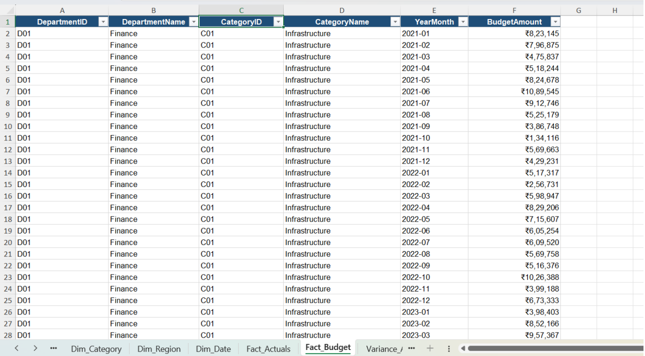
Explanation / Image Caption

The **Fact_Actuals** table represents the granular core of the Star Schema, storing 10,000 actual General Ledger transaction records across 2021–2023:

- **Transaction-Level Detail:** Captures individual expenditure vouchers including TransactionID, DateKey, PaymentMethod (Card, Bank Transfer, Cash, UPI), and ActualAmount.
- **Foreign Key References:** Contains surrogate foreign keys (DepartmentID, CategoryID, RegionID, DateKey, YearMonth) connecting directly to all dimension tables.
- **Data Quality Control:** Incorporates the DataQualityFlag attribute (*OK* vs. *Needs*

Review) to enable automated audit tracking and filtering of flagged entries prior to Power BI reporting.

Target Budget Fact Table (Fact_Budget):



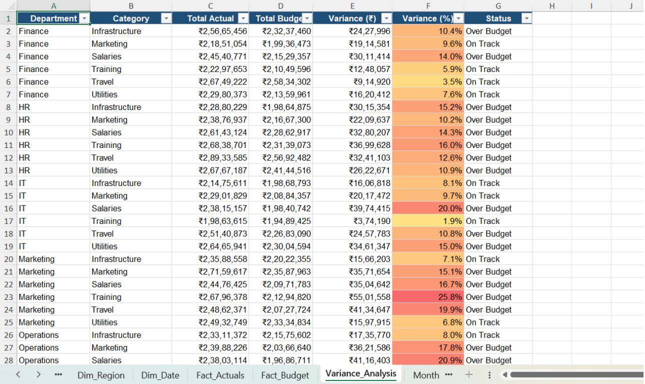
DepartmentID	DepartmentName	CategoryID	CategoryName	YearMonth	BudgetAmount
D01	Finance	C01	Infrastructure	2021-01	₹8,23,145
D01	Finance	C01	Infrastructure	2021-02	₹7,96,875
D01	Finance	C01	Infrastructure	2021-03	₹4,75,837
D01	Finance	C01	Infrastructure	2021-04	₹5,18,244
D01	Finance	C01	Infrastructure	2021-05	₹8,24,678
D01	Finance	C01	Infrastructure	2021-06	₹10,89,545
D01	Finance	C01	Infrastructure	2021-07	₹9,12,746
D01	Finance	C01	Infrastructure	2021-08	₹5,25,179
D01	Finance	C01	Infrastructure	2021-09	₹3,86,748
D01	Finance	C01	Infrastructure	2021-10	₹1,34,116
D01	Finance	C01	Infrastructure	2021-11	₹5,69,663
D01	Finance	C01	Infrastructure	2021-12	₹4,29,231
D01	Finance	C01	Infrastructure	2022-01	₹5,17,317
D01	Finance	C01	Infrastructure	2022-02	₹2,56,731
D01	Finance	C01	Infrastructure	2022-03	₹5,98,947
D01	Finance	C01	Infrastructure	2022-04	₹8,29,206
D01	Finance	C01	Infrastructure	2022-05	₹7,15,607
D01	Finance	C01	Infrastructure	2022-06	₹6,05,254
D01	Finance	C01	Infrastructure	2022-07	₹6,09,520
D01	Finance	C01	Infrastructure	2022-08	₹5,69,758
D01	Finance	C01	Infrastructure	2022-09	₹5,16,376
D01	Finance	C01	Infrastructure	2022-10	₹10,26,388
D01	Finance	C01	Infrastructure	2022-11	₹3,99,188
D01	Finance	C01	Infrastructure	2022-12	₹6,73,333
D01	Finance	C01	Infrastructure	2023-01	₹3,98,403
D01	Finance	C01	Infrastructure	2023-02	₹8,52,166
D01	Finance	C01	Infrastructure	2023-03	₹9,57,367

Explanation:

The Fact_Budget table forms the secondary fact table in the Star Schema, storing pre-aggregated departmental target budgets structured at Department × Category × Month granularity across 2021–2023:

- Departmental Target Granularity: Captures planned expenditure targets (BudgetAmount in ₹) aggregated by monthly financial periods (YearMonth) for each organizational unit and expense classification.
- Foreign Key References: Features relational keys (DepartmentID, CategoryID, and YearMonth) that align seamlessly with Dim_Department, Dim_Category, and Dim_Date.
- Variance Engine Integration: Serves as the baseline comparative dataset for DAX variance calculations, enabling Power BI to evaluate planned targets against transactional actuals from Fact_Actuals across custom date ranges and organizational slices.

Departmental Variance Analysis Matrix (Variance_Analysis):



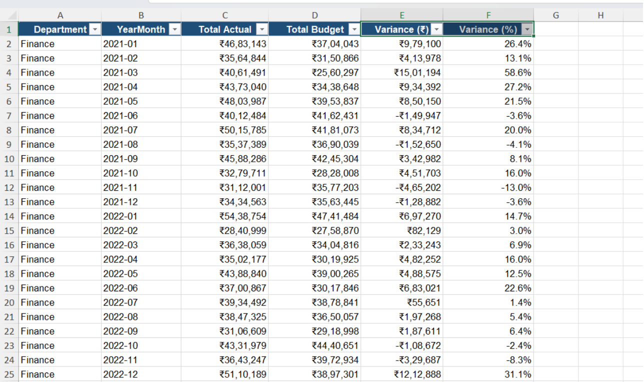
Department	Category	Total Actual	Total Budge	Variance (₹)	Variance (%)	Status
Finance	Infrastructure	₹2,56,65,456	₹2,32,37,460	₹24,27,996	10.4%	Over Budget
Finance	Marketing	₹2,18,51,054	₹1,99,36,473	₹19,14,581	9.6%	On Track
Finance	Salaries	₹2,45,40,771	₹2,15,29,357	₹30,11,414	14.0%	Over Budget
Finance	Training	₹2,22,97,653	₹2,10,49,596	₹12,48,057	5.9%	On Track
Finance	Travel	₹2,67,49,222	₹2,58,34,302	₹9,14,920	3.5%	On Track
Finance	Utilities	₹2,29,80,373	₹2,13,59,961	₹16,20,412	7.6%	On Track
HR	Infrastructure	₹2,28,80,229	₹1,98,64,875	₹30,15,354	15.2%	Over Budget
HR	Marketing	₹2,38,76,937	₹2,16,67,300	₹22,09,637	10.2%	Over Budget
HR	Salaries	₹2,61,43,124	₹2,28,62,917	₹32,80,207	14.3%	Over Budget
HR	Training	₹2,68,38,701	₹2,31,39,073	₹36,99,628	16.0%	Over Budget
HR	Travel	₹2,89,33,585	₹2,56,92,482	₹32,41,103	12.6%	Over Budget
HR	Utilities	₹2,67,67,187	₹2,41,44,516	₹26,22,671	10.9%	Over Budget
IT	Infrastructure	₹2,14,75,611	₹1,98,68,793	₹16,06,818	8.1%	On Track
IT	Marketing	₹2,29,01,829	₹2,08,84,357	₹20,17,472	9.7%	On Track
IT	Salaries	₹2,38,15,157	₹1,98,40,742	₹39,74,415	20.0%	Over Budget
IT	Training	₹1,98,63,615	₹1,94,89,425	₹3,74,190	1.9%	On Track
IT	Travel	₹2,51,40,873	₹2,26,83,090	₹24,57,783	10.8%	Over Budget
IT	Utilities	₹2,64,65,941	₹2,30,04,594	₹34,61,347	15.0%	Over Budget
Marketing	Infrastructure	₹2,35,88,558	₹2,20,22,355	₹15,66,203	7.1%	On Track
Marketing	Marketing	₹2,71,59,617	₹2,35,87,963	₹35,71,654	15.1%	Over Budget
Marketing	Salaries	₹2,44,76,425	₹2,09,71,783	₹35,04,642	16.7%	Over Budget
Marketing	Training	₹2,67,96,378	₹2,12,94,820	₹55,01,558	25.8%	Over Budget
Marketing	Travel	₹2,48,62,371	₹2,07,27,724	₹41,34,647	19.9%	Over Budget
Marketing	Utilities	₹2,49,32,749	₹2,33,34,834	₹15,97,915	6.8%	On Track
Operations	Infrastructure	₹2,33,11,372	₹2,15,75,602	₹17,35,770	8.0%	On Track
Operations	Marketing	₹2,39,88,226	₹2,03,66,640	₹36,21,586	17.8%	Over Budget
Operations	Salaries	₹2,38,03,114	₹1,96,86,711	₹41,16,403	20.9%	Over Budget

Explanation:

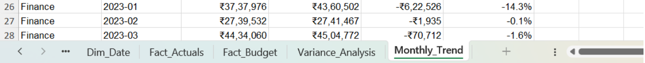
The Variance_Analysis sheet provides a consolidated performance matrix evaluating actual expenditures against baseline budgets across all Department × Category combinations:

- **Comparative Financial Metrics:** Displays aggregated metrics including Total Actual, Total Budget, Variance (₹), and Variance (%) computed across each expense line item in Indian Rupees.
- **Conditional Thresholds & Status Flags:** Applies automated logic to categorize line items into operational status tiers (e.g., Over Budget vs. On Track) based on variance thresholds, making it easy to isolate high-risk cost drivers like Marketing Training (+25.8% / **+₹55,01,558**) and Operations Salaries (+20.9% / **+₹41,16,403**).

Departmental Monthly Expenditure & Variance Trend (Monthly_Trend):



Department	YearMonth	Total Actual	Total Budget	Variance (₹)	Variance (%)
Finance	2021-01	₹46,83,143	₹37,04,043	₹9,79,100	26.4%
Finance	2021-02	₹35,64,844	₹31,50,866	₹4,13,978	13.1%
Finance	2021-03	₹40,61,491	₹25,60,297	₹15,01,194	58.6%
Finance	2021-04	₹43,73,040	₹34,38,648	₹9,34,392	27.2%
Finance	2021-05	₹48,03,987	₹39,53,837	₹8,50,150	21.5%
Finance	2021-06	₹40,12,484	₹41,62,431	-₹1,49,947	-3.6%
Finance	2021-07	₹50,15,785	₹41,81,073	₹8,34,712	20.0%
Finance	2021-08	₹35,37,389	₹36,90,039	-₹1,52,650	-4.1%
Finance	2021-09	₹45,88,286	₹42,45,304	₹3,42,982	8.1%
Finance	2021-10	₹32,79,711	₹28,28,008	₹4,51,703	16.0%
Finance	2021-11	₹31,12,001	₹35,77,203	-₹4,65,202	-13.0%
Finance	2021-12	₹34,34,563	₹35,63,445	-₹1,28,882	-3.6%
Finance	2022-01	₹54,38,754	₹47,41,484	₹6,97,270	14.7%
Finance	2022-02	₹28,40,999	₹27,58,870	₹82,129	3.0%
Finance	2022-03	₹36,38,059	₹34,04,816	₹2,33,243	6.9%
Finance	2022-04	₹35,02,177	₹30,19,925	₹4,82,252	16.0%
Finance	2022-05	₹43,88,840	₹39,00,265	₹4,88,575	12.5%
Finance	2022-06	₹37,00,867	₹30,17,846	₹6,83,021	22.6%
Finance	2022-07	₹39,34,492	₹38,78,841	₹55,651	1.4%
Finance	2022-08	₹38,47,325	₹36,50,057	₹1,97,268	5.4%
Finance	2022-09	₹31,06,609	₹29,18,998	₹1,87,611	6.4%
Finance	2022-10	₹43,31,979	₹44,40,651	-₹1,08,672	-2.4%
Finance	2022-11	₹36,43,247	₹39,72,934	-₹3,29,687	-8.3%
Finance	2022-12	₹51,10,189	₹38,97,301	₹12,12,888	31.1%



26	Finance	2023-01	₹37,37,976	₹43,60,502	-₹6,22,526	-14.3%
27	Finance	2023-02	₹27,39,532	₹27,41,467	-₹1,935	-0.1%
28	Finance	2023-03	₹44,34,060	₹45,04,772	-₹70,712	-1.6%

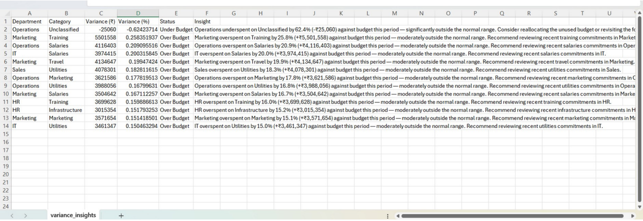
Explanation:

The Monthly_Trend summary sheet provides time-series tracking of financial actuals against budget baselines across individual monthly periods (YearMonth) from 2021 through 2023:

- Temporal Performance Granularity: Breaks down expenditures by department across continuous monthly timelines in Indian Rupees (₹), capturing seasonal swings, year-end spend surges (e.g., Finance December 2022 over-budget spike of +31.1% / ₹12,12,888), and fiscal year shifts.
- Trend & Time-Intelligence Foundation: Serves as the pre-aggregated comparative structure used to validate Power BI DAX time-intelligence functions—including Month-over-Month (MoM) growth calculations, Year-to-Date (YTD) actual tracking, and rolling budget variance trends.

Financial_Variance_Data_Model Link: https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Financial_Variance_Data_Model.xlsx

AI-Generated Automated Variance Insights Dataset (variance_insights):



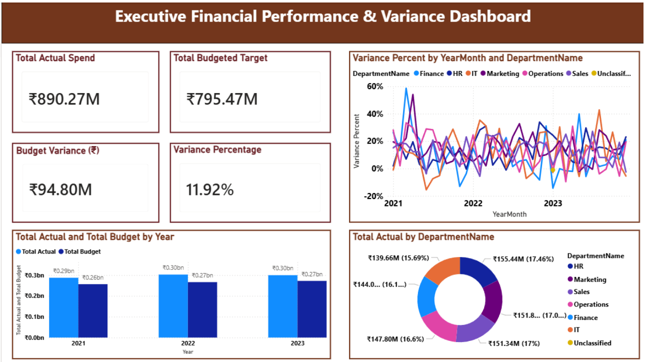
	A	B	C	D	E	F
1	Department	Category	Variance (₹)	Variance (%)	Status	Insight
2	Operations	Unclassified	-25060	-0.62423714	Under Budget	Operations underspent on Unclassified by 62.4% (-₹25,060) against budget this period — significantly outside the normal range. Consider reallocating the unused budget or revisiting the f
3	Marketing	Training	5501558	0.258351937	Over Budget	Marketing overspent on Training by 25.8% (+₹5,501,558) against budget this period — moderately outside the normal range. Recommend reviewing recent training commitments in Market
4	Operations	Salaries	4116403	0.209095516	Over Budget	Operations overspent on Salaries by 20.9% (+₹4,116,403) against budget this period — moderately outside the normal range. Recommend reviewing recent salaries commitments in Oper
5	IT	Salaries	3974415	0.200315845	Over Budget	IT overspent on Salaries by 20.0% (+₹3,974,415) against budget this period — moderately outside the normal range. Recommend reviewing recent salaries commitments in IT.
6	Marketing	Travel	4134647	0.19947424	Over Budget	Marketing overspent on Travel by 19.9% (+₹4,134,647) against budget this period — moderately outside the normal range. Recommend reviewing recent travel commitments in Marketing.
7	Sales	Utilities	4078301	0.182811615	Over Budget	Sales overspent on Utilities by 18.3% (+₹4,078,301) against budget this period — moderately outside the normal range. Recommend reviewing recent utilities commitments in Sales.
8	Operations	Marketing	3621586	0.177819513	Over Budget	Operations overspent on Marketing by 17.8% (+₹3,621,586) against budget this period — moderately outside the normal range. Recommend reviewing recent marketing commitments in
9	Operations	Utilities	3988056	0.16799631	Over Budget	Operations overspent on Utilities by 16.8% (+₹3,988,056) against budget this period — moderately outside the normal range. Recommend reviewing recent utilities commitments in Opera
10	Marketing	Salaries	3504642	0.167112257	Over Budget	Marketing overspent on Salaries by 16.7% (+₹3,504,642) against budget this period — moderately outside the normal range. Recommend reviewing recent salaries commitments in Marke
11	HR	Training	3699628	0.159886613	Over Budget	HR overspent on Training by 16.0% (+₹3,699,628) against budget this period — moderately outside the normal range. Recommend reviewing recent training commitments in HR.
12	HR	Infrastructure	3015354	0.151793253	Over Budget	HR overspent on Infrastructure by 15.2% (+₹3,015,354) against budget this period — moderately outside the normal range. Recommend reviewing recent infrastructure commitments in H
13	Marketing	Marketing	3571654	0.151418501	Over Budget	Marketing overspent on Marketing by 15.1% (+₹3,571,654) against budget this period — moderately outside the normal range. Recommend reviewing recent marketing commitments in Ma
14	IT	Utilities	3461347	0.150463294	Over Budget	IT overspent on Utilities by 15.0% (+₹3,461,347) against budget this period — moderately outside the normal range. Recommend reviewing recent utilities commitments in IT.

Explanation: The variance_insights dataset serves as the underlying feed for the project's AI Narrative Layer, storing automated natural-language summaries generated for key variance anomalies:

- **Natural-Language Insight Structure:** Maps departmental cost deviations (**Variance (₹)** and **Variance (%)**) to structured textual narratives (Insight column) detailing spend behavior, anomaly severity, and actionable management recommendations.
- **Executive Actionability:** Captures specific operational variance drivers—such as Operations Unclassified underspend (**-62.4% / -₹25,060**) and Marketing Training overspend (**+25.8% / +₹55,01,558**)—providing narrative commentary that management can read alongside the Power BI dashboard.

variance_insights Link: https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/variance_insights.csv

POWER BI DASHBOARD:



Executive Financial Performance & Variance Dashboard

Total Actual Spend
₹890.27M

Total Budgeted Target
₹795.47M

Budget Variance (₹)
₹94.80M

Variance Percentage
11.92%

Explanation:

This dashboard represents the final visualization layer of the project, consolidating the star-schema data model (Fact_Actuals, Fact_Budget, and supporting dimension tables) into a single executive view for budget variance monitoring:

- **KPI Summary Cards:** Displays the four headline metrics — Total Actual Spend (₹890.27M), Total Budgeted Target (₹795.47M), Budget Variance (₹94.80M), and Variance Percentage (11.92%) — giving an at-a-glance view of overall financial performance against plan for the 2021–2023 period.
- **Total Actual and Total Budget by Year:** A year-over-year comparison bar chart showing that actual spend exceeded budget in every year (2021, 2022, and 2023), indicating a consistent, structural overspend pattern rather than a single anomalous period.
- **Total Actual by DepartmentName:** A donut chart breaking down total actual expenditure across all seven departments, showing HR (17.46%), Marketing, Sales, Operations, Finance, and IT as the primary cost centers, with Unclassified (0.01%) representing data-quality-flagged transactions.
- **Variance Percent by YearMonth and DepartmentName:** A time-series line chart tracking each department's variance percentage on a monthly basis, enabling identification of when and where budget deviations occurred. The most significant anomaly is Finance's spike to approximately 58% in March 2021 (Finance, +58.6%), which is also visible in the Monthly_Trend sheet.

Power BI Dashboard Link:
https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Executive%20Financial%20Performance%20%26%20Variance%20Dashboard.pbix

AI-Generated Variance Insights:

Marketing – Training (+25.8% Over Budget): Marketing recorded the highest overrun, spending ₹55,01,558 (+25.8%) above budget. A review of recent training vendor commitments is recommended.

Operations – Salaries (+20.9% Over Budget): Operations exceeded its baseline salary budget by ₹41,16,403 (+20.9%).

IT – Salaries (+20.0% Over Budget): IT departmental salary expenditures ran ₹39,74,415 (+20.0%) higher than planned.

Marketing – Travel (+19.9% Over Budget): Marketing travel spend exceeded budget by ₹41,34,647 (+19.9%). An audit of recent travel bookings is suggested.

Sales – Utilities (+18.3% Over Budget): Sales exceeded its utility budget allocation by ₹40,78,301 (+18.3%).

HR – Training (+16.0% Over Budget): HR overspent on training initiatives by ₹36,99,628 (+16.0%) against baseline targets.

Operations – Unclassified (-62.4% Under Budget): Operations underspent on Unclassified line items by ₹25,060 (-62.4%). Consider reallocating these unutilized funds or updating forecast models.

Conclusion:

This project built a complete AI-powered budget variance pipeline — from raw Tally-style transaction data, through a cleaned star-schema model in Excel, to an interactive Power BI dashboard with AI-generated insights. The analysis found the organization ran **11.92% over budget (₹94.80M)** consistently across 2021–2023, with **Finance** showing the most volatile variance and **HR, Marketing, and Sales** as the top spending departments. The AI Insight Layer automatically explained the biggest variances in plain language, cutting down the manual reporting effort a finance team would otherwise need. Overall, the project meets its goal: automating variance detection and making it easy to see *what* went over budget, *where*, and *when*.

THANK YOU